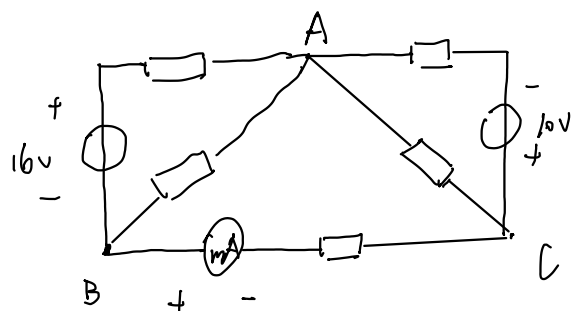
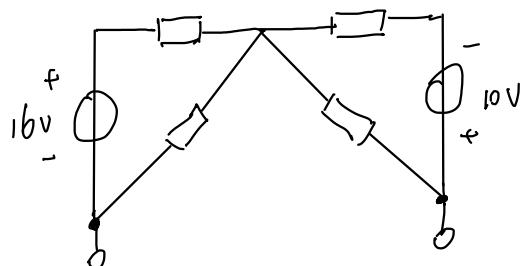


叠加定理:

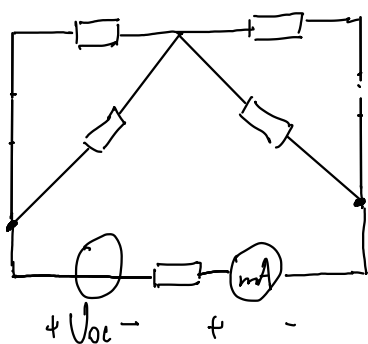


U_{AB} , U_{AC} , U_{BC} , I_{BC}

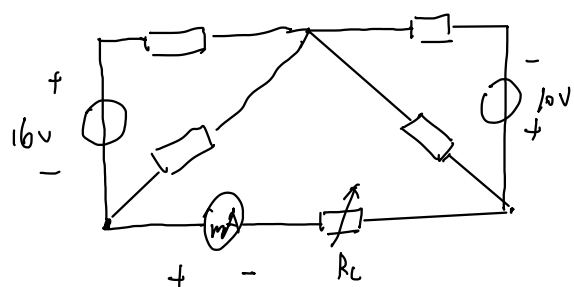
戴维南定理:



U_{oc} , $I_{sc} \Rightarrow R_i$



$\Rightarrow I'$, 与 I_{BL} 比较



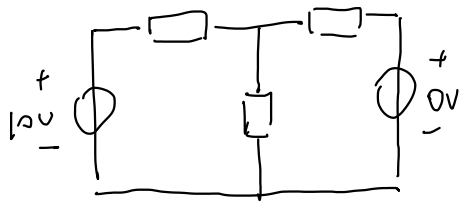
$R_L: 50 \sim 250 \Omega$

$\downarrow$
 U_R, I_R, P

$\Rightarrow U_R - I_R$ 曲线

与理论曲线 $U = U_{oc} - I R_i$

特勒根定理:

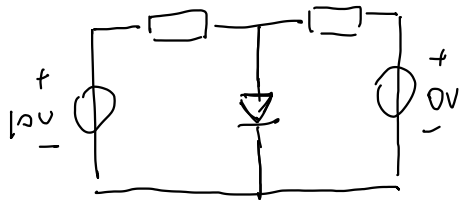


$$R_1, R_2, R_3, U_{S1}, U_{S2}$$

$$U_k$$

$$i_k$$

$$\Rightarrow \sum U_k i_k = 0$$



$$R_1, R_2, \text{二极管}, U_{S1}, U_{S2}$$

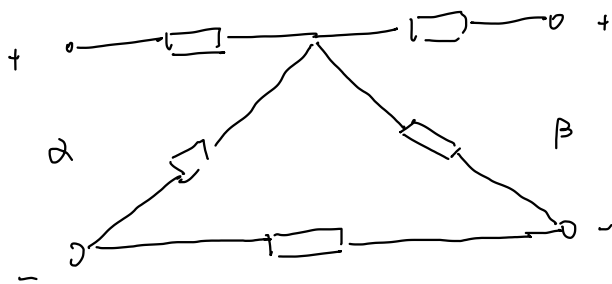
$$\hat{U}_k$$

$$\hat{i}_k$$

$$\Rightarrow \sum U_k \hat{i}_k = 0$$

$$\sum \hat{U}_k \hat{i}_k = 0$$

互易定理:



$$U_{S\alpha} \rightarrow i_\beta$$

$$\hat{U}_{S\beta} \rightarrow \hat{i}_\alpha$$

$\Downarrow$

$$\frac{U_{S\alpha}}{i_\beta} = \frac{\hat{U}_{S\beta}}{\hat{i}_\alpha}$$

$$i_{S\alpha} \rightarrow U_\beta$$

$$\hat{i}_{S\beta} \rightarrow U_\alpha$$

$\Downarrow$

$$\frac{i_{S\alpha}}{U_\beta} = \frac{\hat{i}_{S\beta}}{U_\alpha}$$

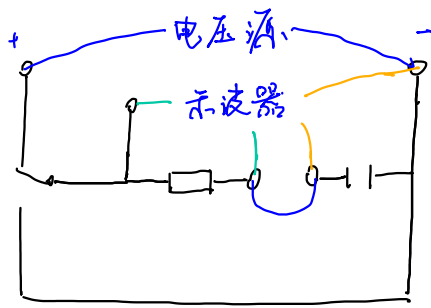
$$i_{S\alpha} \rightarrow i_\beta$$

$$\hat{U}_{S\beta} \rightarrow \hat{U}_\alpha$$

$\Downarrow$

$$\frac{i_{S\alpha}}{i_\beta} = \frac{\hat{U}_{S\beta}}{\hat{U}_\alpha}$$

一阶电路：



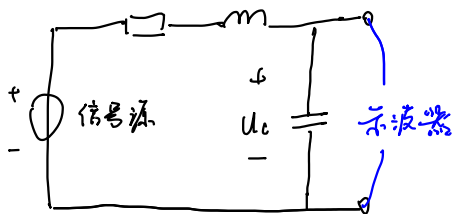
$$\tau = RC$$

$$\Rightarrow t = \tau \text{ 时 } U_R, U_C$$

(理论: 0.632, 0.368 倍)

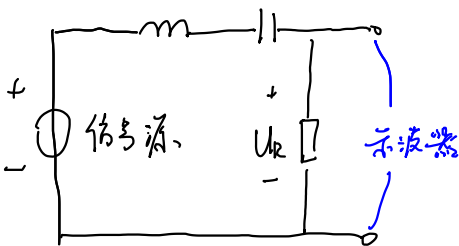
$\Rightarrow$ 4 幅曲线.

二阶电路：



(由 + 到 - : R, L, C)

$C = 0.01 \mu F$, $L = 0.01 H$, $\pm 8V$ 方波
 $\downarrow$
 1kHz 左右

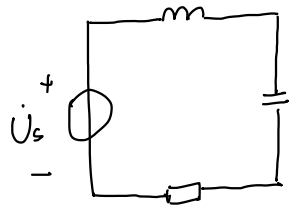


过阻尼: $R > 2\sqrt{\frac{L}{C}}$

临界阻尼: $R = 2\sqrt{\frac{L}{C}} = 2k\Omega \rightarrow \alpha = \omega_0 = \frac{1}{\sqrt{LC}}$

欠阻尼: $R < 2\sqrt{\frac{L}{C}} \rightarrow \text{测 } T_d, \omega_d = \frac{\omega_0}{T_d}$

RLC 谐振:



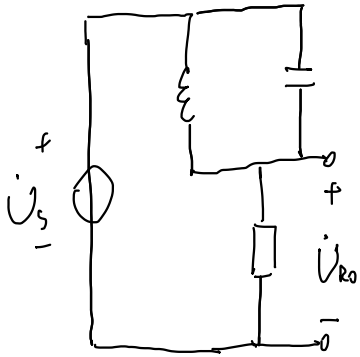
$$R = 400\Omega / 1k\Omega \quad L = 0.1H, \quad C = 0.01\mu F, \quad U_s = 1V$$

$$f: 4.0 \sim 6.0 \text{ kHz}$$

测 $U_R, U_L, U_C \rightarrow f_0$ (U_R 最大时)

计算 $f/f_0, I/I_0$, 绘制曲线; 绘制 $U_L, U_C - \omega$ 关系曲线

$$\text{计算 } Q_{\text{理论}} = \frac{1}{R+R_L} \sqrt{\frac{L}{C}}, \quad Q_{\text{实验}} = \frac{2\pi f_0 L}{R+R_0}$$



$$R_0 = 100\Omega \quad L = 0.01H, \quad C = 0.1\mu F, \quad U_s = 1V$$

$$f: 4.1 \sim 6.1 \text{ kHz}$$

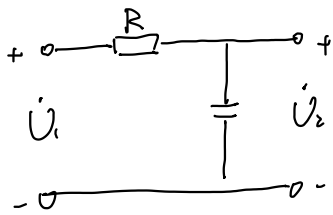
测 $U_{R_0} \rightarrow f_0$ (U_{R_0} 最大时)

计算 $f/f_0, |Z|/Z_0$, 绘制曲线

$$\text{计算 } Q_{\text{理论}} = \frac{1}{R} \sqrt{\frac{L}{C}}, \quad Q_{\text{实验}} = \frac{2\pi f_0 L}{R}$$

频率特性:

低通 (电容出)



$$U_1 = 1V, R = 8k\Omega, C = 0.01\mu F$$

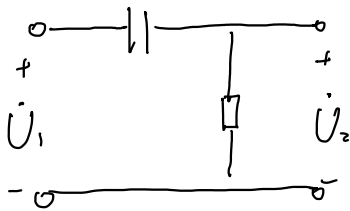
$$f: 200Hz \sim 10kHz$$

$$\text{调 } U_2 \rightarrow f_0 (U_2 = 0.707 U_1 \text{ 时})$$

绘制 $U_2 - f$ 曲线 (半对数纸)

$$\text{计算 } f_{0\text{理论}} = \frac{1}{2\pi RC}$$

高通 (电阻出)



$$U_1 = 1V, R = 4k\Omega, C = 0.1\mu F$$

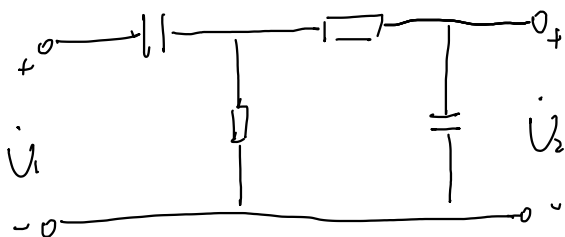
$$f: 100Hz \sim 10kHz$$

$$\text{调 } U_2 \rightarrow f_0 (U_2 = 0.707 U_1 \text{ 时})$$

绘制 $U_2 - f$ 曲线 (半对数纸)

$$\text{计算 } f_{0\text{理论}} = \frac{1}{2\pi RC}$$

带通 (高通 + 低通)



$$U_1 = 1V$$

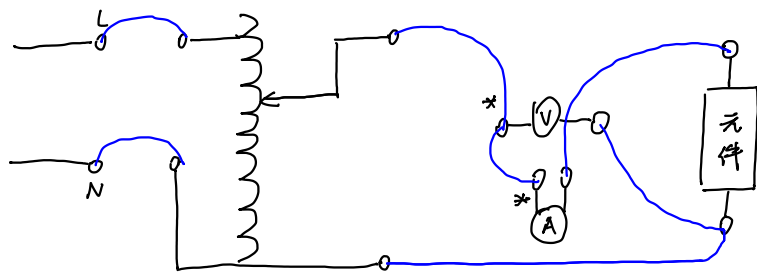
$$f: 50Hz \sim 10kHz$$

$$\text{调 } U_2 \rightarrow f_0 (U_2 \text{ max 时})$$

$$f_1, f_2 (U_2 = 0.707 U_{2\text{max}} \text{ 时})$$

绘制 $U_2 - f$ 曲线 (半对数纸)

交流参数:



R: 不同 I (0.5A, 0.7A, 0.9A), 测 U , P , 计算 $R = \frac{U}{I}$, $R = \frac{U^2}{P}$

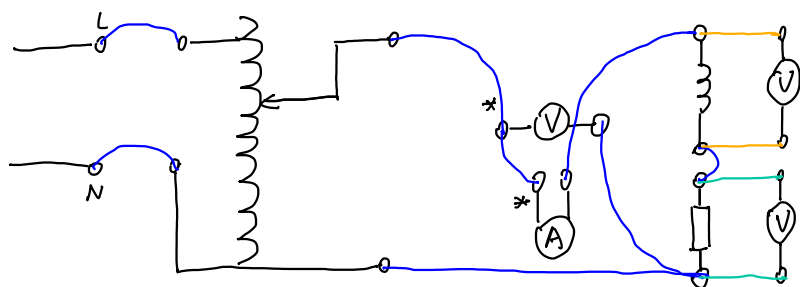
C: 不同 U (160V, 180V, 200V), 测 I , P , 计算 $C = \frac{I}{\omega U} = \frac{I}{2\pi f U}$

L: 不同 I (0.5A, 0.7A, 0.9A), 测 U , P , 计算 $R_L = \frac{P}{I^2}$, $L = \frac{Z_L}{\omega} = \frac{\sqrt{|Z_L|^2 - R_L^2}}{2\pi f} = \frac{\sqrt{(\frac{U}{I})^2 - R_L^2}}{2\pi f}$

R, C, L 串联: 不同 I (0.5A, 0.7A, 0.9A), 测 P , U , U_R , U_C , U_L , 计算 $Z = R + jX$,
分别作相量图.

(其中, $R = \frac{P}{I^2}$, $X = \sqrt{|Z|^2 - R^2} = \sqrt{(\frac{U}{I})^2 - R^2}$)

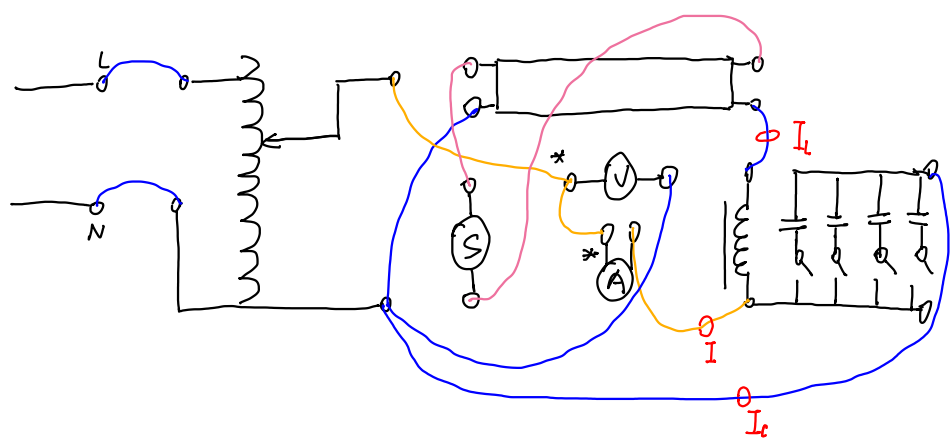
三电压表法



不同 I (0.5A, 0.7A, 0.9A), 测 U , U_L , U_R , 计算 $R_L = \frac{U_L}{I} \cos \varphi_L$, $L = \frac{\sqrt{(\frac{U_L}{I})^2 - R_L^2}}{2\pi f}$

(其中, $\cos \varphi_L = \frac{U^2 - U_R^2 - U_L^2}{2 U_R \cdot U_L}$)

提高功率因数：

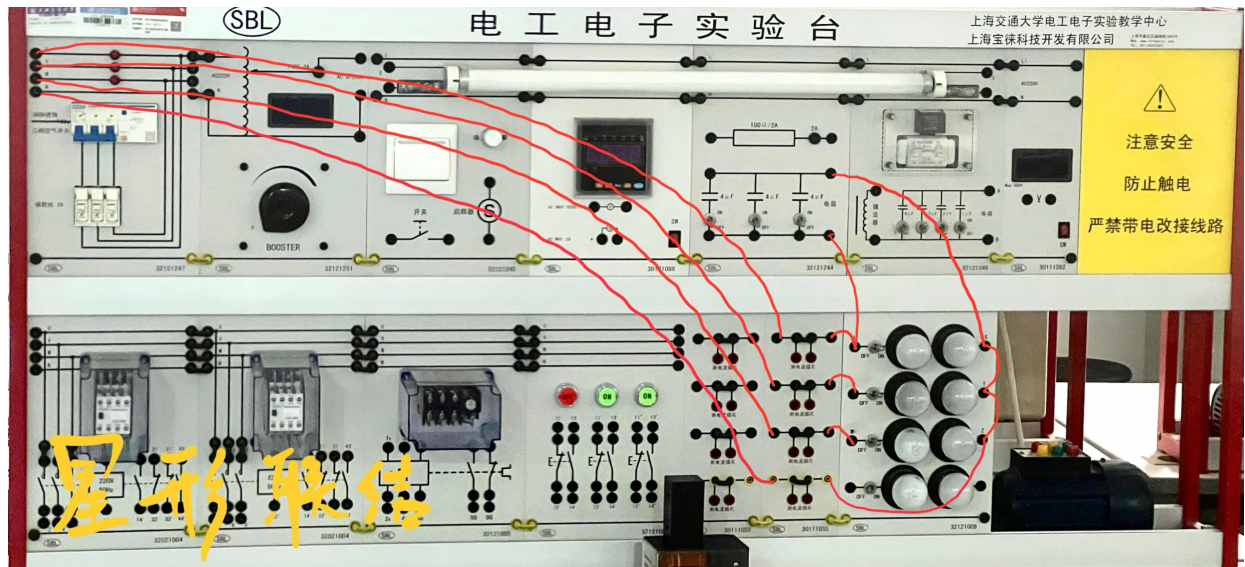


$C = 0 \mu F \rightarrow$ 测 I, U, P, U_R, U_L , 计算 $\cos\varphi = \frac{P}{UI}, R = \frac{P}{I^2}, L = \frac{\sqrt{(\frac{U_L}{I})^2 - R^2}}{2\pi f}$

$C = 1 \sim 7 \mu F \rightarrow$ 测 I, I_L, I_C, P , 计算 $\cos\varphi = \frac{P}{UI}, \varphi$

绘制 $\cos\varphi - C, I - C$ 关系曲线。

三相电路：



测相序：U 亮 $\rightarrow$ V 亮 $\rightarrow$ W 暗

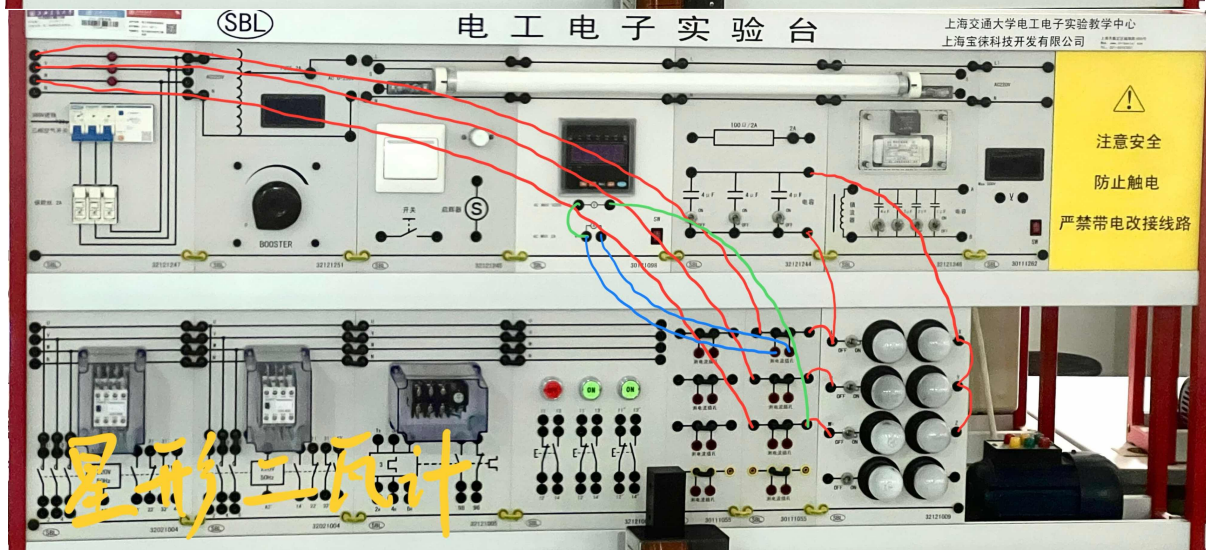
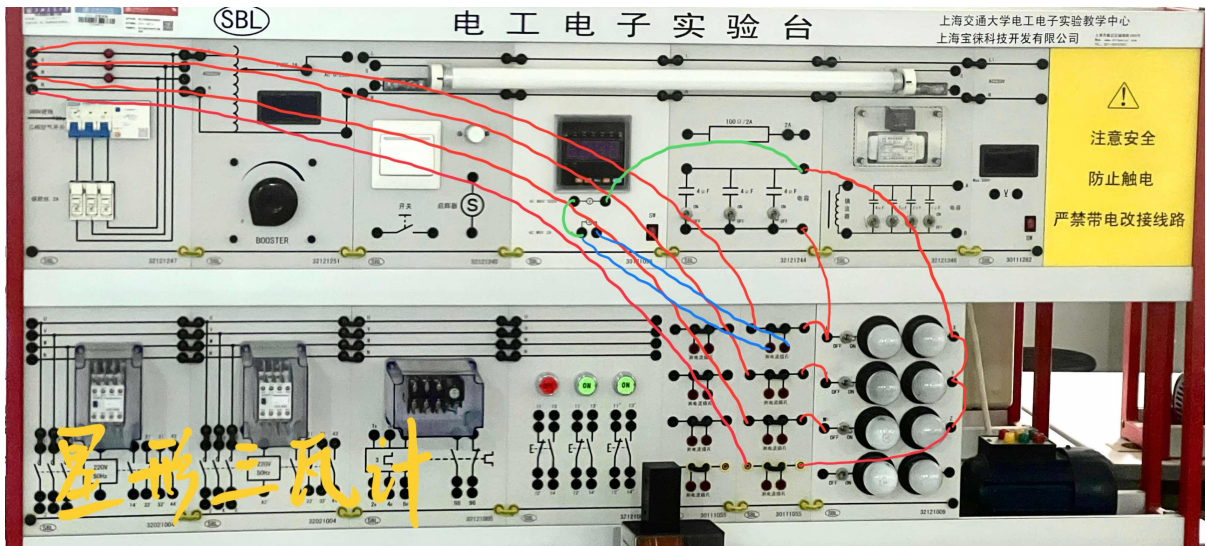
四线制：U 相：电容、4mF 电容、开路

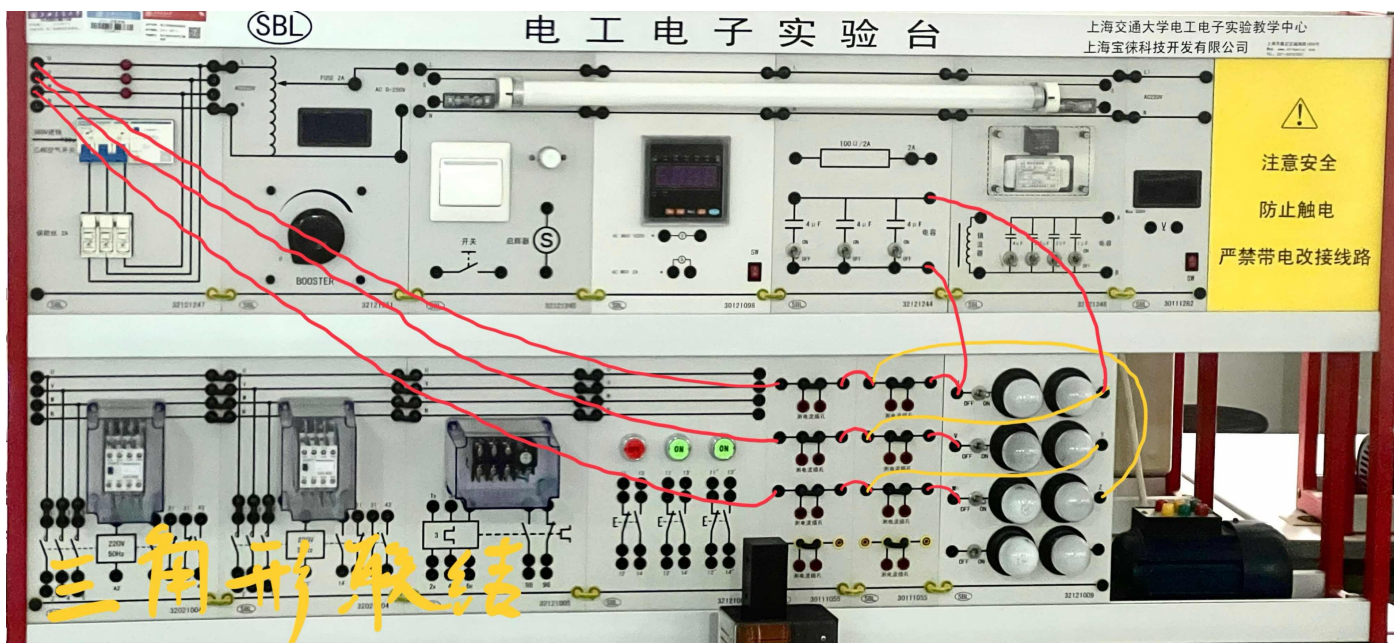
测 线电压、相电压、中线电压、线电流、中线电流

三线制：U 相：电容、4mF 电容、开路

测 相电压、中线电压、线电流

原相量图





U相: 电容、4mF电容、开路

测线电流、相电流

画相量图

